

Beyond Reported Scores: Constructing an ESG Misrepresentation Index for the Tobacco Sector

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Note: This document contains a selected excerpt of the full report. If you would like to read the complete analysis, findings, and supporting visualisations, feel free to get in touch; I'd be happy to share the complete version upon request.

1. Introduction

This report examines whether British American Tobacco's ESG ratings genuinely reflect its real-world impact, or whether they overstate progress in an industry whose core product remains harmful by design. The study argues that conventional ESG frameworks often struggle in controversial sectors because they reward disclosure quality, operational improvements, and relatively easier-to-measure changes, while underrepresenting product-level harm, youth recruitment, and long-term public-health effects. To explore this issue, the report develops an Analytic Hierarchy Process (AHP) weighting structure and an ESG Misrepresentation Index (EMI), then compares BAT's reported ESG score with a materially adjusted score from 2023 to 2025. The results show a clear and widening gap between reported performance and materially grounded performance, suggesting that BAT's ESG narrative is improving faster than its actual harm profile.

The report asks a focused question: to what extent do BAT's ESG ratings capture genuine sustainability progress, given evidence that the company may be shifting harm rather than reducing it? BAT's "A Better Tomorrow" strategy and its New Generation Products are presented as evidence of transition, but the report questions whether this represents substantive harm reduction or simply a change in product mix and messaging. This distinction matters because a company can improve its ESG appearance without materially reducing the damage created by its core business. The EMI is designed to make that gap visible in a structured and measurable way.

2. Brief Literature Context

The literature review shows that ESG ratings are often imperfect proxies for real corporate impact. One major issue is rating divergence: different agencies frequently produce very different scores for the same company because their methods, weighting choices, and definitions of materiality differ. Another problem is the boundary issue, where standard

ESG frameworks fail to capture harms that sit outside conventional operational metrics, especially in tobacco where the main harm comes from the product itself rather than from emissions or internal governance alone. A third concern is ESG arbitrage, where firms can improve their standing within rating systems by strengthening disclosures or operational processes without reducing the underlying social damage. In a sector like tobacco, this creates a risk that surface-level progress is mistaken for meaningful change.

3. Data Section

The report uses ESG ratings from MSCI, Sustainalytics, CDP, ISS ESG, and S&P CSA, along with BAT annual reports, sustainability disclosures, health datasets, and geographic production data. Scores were normalised to a common scale and then combined into a reported ESG composite. The AHP framework assigns greater weight to Social factors, especially youth vaping, dependency, and geographic harm transfer, because these are the most direct indicators of tobacco-related harm. The EMI is then defined as the gap between the reported ESG score and the materially adjusted AHP-based score.

Table 1: Table 1. AHP Structure

Pillar	Weight	Main Sub-Criteria	Rationale
Environmental (E)	0.25	Cigarette butts, vape waste, deforestation	Important, but less direct than product harm
Social (S)	0.5	Youth vaping, dependency, Sub-Saharan Africa (SSA) shift	Closest to the company's real harm footprint
Governance (G)	0.25	Disclosure, lobbying, incentives	Shapes transparency and risk management

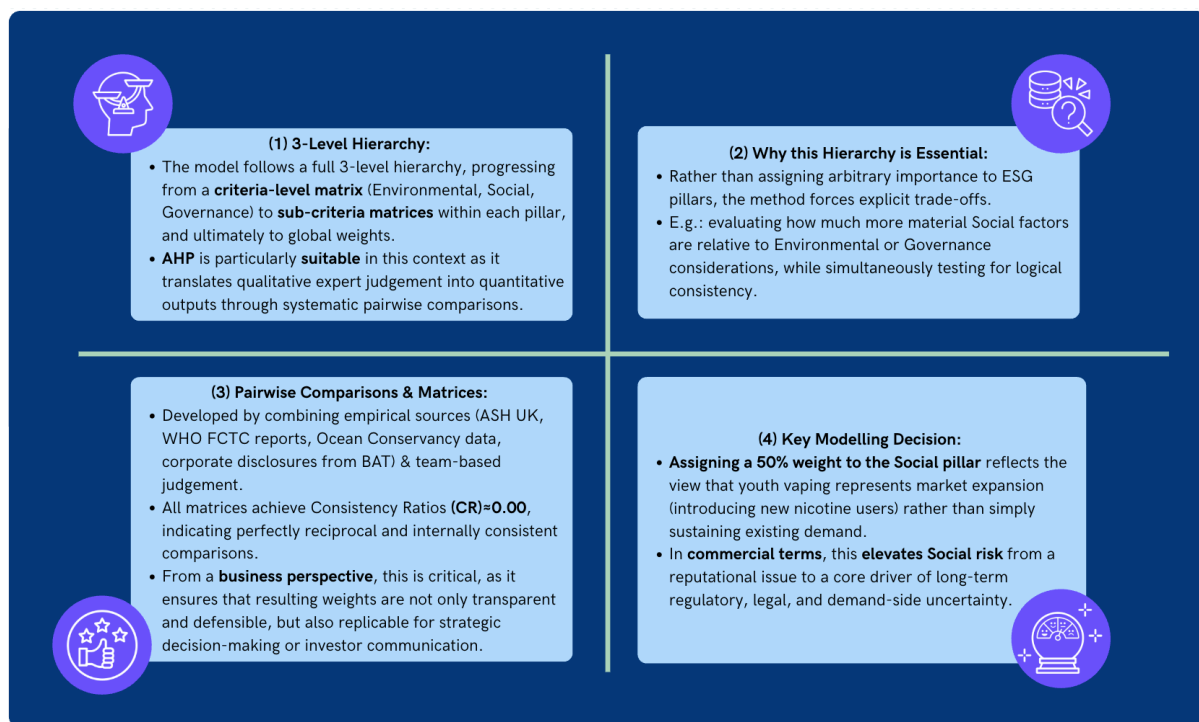
This weighting scheme is important because it shifts the focus from what is easiest to report to what is most consequential. Rather than treating all ESG pillars as equally

informative, it reflects the idea that some harms in tobacco are much more material than others. That is the conceptual basis of the EMI framework.

4. Methodology

4.1. Analytic Hierarchy Process (AHP) ([Link to Dashboard](#))

This study applies the AHP to construct a structured, evidence-based ESG weighting framework.

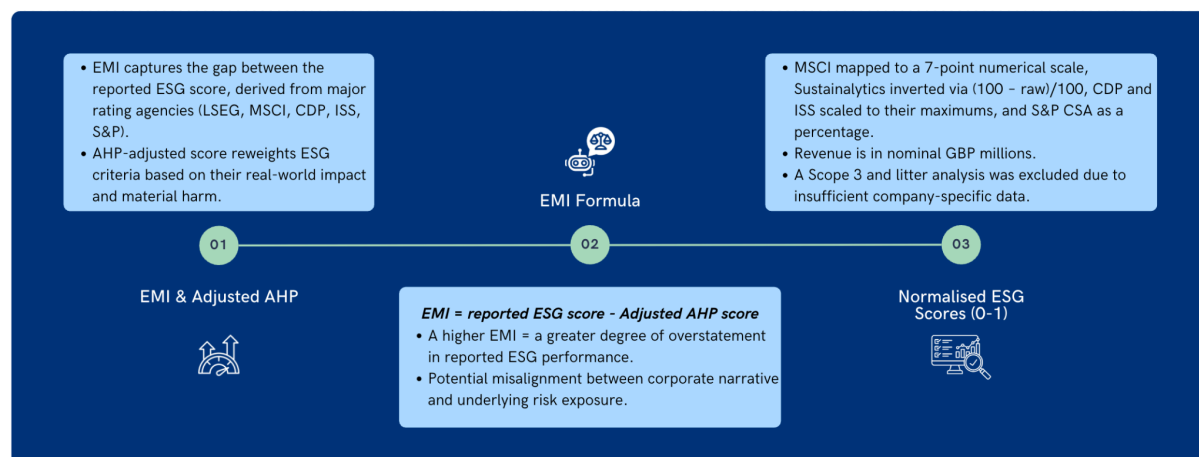


Diag.1 AHP Framework processes

4.2. Gap Analysis: ESG Misrepresentation Index (EMI)

The ESG Misrepresentation Index compares BAT's externally reported ESG score with a materially adjusted score derived from the AHP framework. The reported score reflects agency ratings after normalisation, while the adjusted score reflects evidence-based judgments about the severity and persistence of harms. A larger EMI means a larger gap

between how the company is presented and how it performs under a more critical, impact-focused lens. The purpose is not to claim a single perfect score, but to show whether the direction of change in reported ESG aligns with the direction of change in material harm.



Diag. 2 Explanations of EMI adjusted-AHP, and normalisation of ESG scores

Table 2: Reported ESG vs AHP-Adjusted ESG

Year	Reported ESG	AHP-Adjusted Score	EMI (Gap)	Interpretation
2023	74.5	35.0	39.5	Large initial gap
2024	77.8	37.0	40.8	Gap widens
2025	81.0	39.0	42.0	EMI up by 2.5 pts

The pattern is revealing. BAT's reported ESG score increases by 6.5 points from 2023 to 2025, while the AHP-adjusted score increases by only 4 points over the same period. As a result, the EMI gap widens, which suggests that the company's external ESG image is improving faster than the underlying impact being measured by the more critical framework. The largest weakness remains in the Social pillar, especially youth vaping and the continued dependence on nicotine markets.

4.3. Forecast Analysis ([Link to Dashboard](#))

Table 3: Forecast Highlights

Analysis	Key Result	Interpretation
Smokeless consumer forecast	Strong growth to 2030	Indicates commercial transition
Emissions trend	Downward path for Scope 1 and 2	Shows operational decarbonisation
Revenue crossover	Smokeless may overtake combustibles in the mid-2030s	Suggests structural business change
Transition margin	Growth cushion remains narrow	Strategy may still be fragile

The forecast analysis combined several forward-looking models to assess whether BAT's transition toward smokeless products and lower emissions is likely to translate into a meaningful business shift. First, a Holt-Winters time-series forecast was used to project the growth of smokeless consumers, suggesting a steady increase from 37.5 million in 2026 to 53.1 million by 2030 and 72.6 million by 2035, which indicates that BAT's new-category consumer base is likely to keep expanding if current trends continue. Second, a linear emissions forecast showed a consistent decline in Scope 1 and 2 emissions, with an estimated reduction of about 55.7 ktCOe per year and a projected path toward near-zero emissions by 2029, although this rests on a linear assumption that may become less realistic over time.

Third, a revenue crossover model estimated that smokeless revenue could overtake combustibles around 2039, highlighting a structural shift in the business model but also showing that the timing remains sensitive to small changes in growth rates. Finally, an optimisation model suggested that BAT would need a smokeless growth rate of about 13.3% to maintain a £27 billion revenue floor by 2030, while the observed smokeless CAGR



was already close to that threshold, leaving only a narrow margin for error. Taken together, these forecasts suggest that BAT's transition is commercially meaningful but still fragile, and that rising smokeless adoption should not automatically be interpreted as evidence of reduced societal harm.

5. Results

The AHP-EMI framework shifts the analysis from reporting ESG scores to materially weighted outcomes. The results are presented across 4 stages.

5.1. Why Mainstream Frameworks Overrate BAT

Fewer than 37% of LSEG's score comes from outcome and impact measures, with the remainder driven by policies and processes (Guerrero & Viteri, 2025). Materiality weights are derived from peer-group disclosure averages (LSEG, 2024), meaning that tobacco companies are benchmarked against each other rather than absolute harm. BAT's MSCI A rating against PMI's BBB, despite near identical portfolios, reflect superior disclosure, but not societal outcomes (Morningstar, 2021). The Sustainalytics improvement from 32.8 to 21.0 represents enhanced reporting rather than minimised societal harm (Sustainalytics, 2020). Importantly, the three dimensions that are most underweighted represent 47.5% of AHP weight.

5.2. How the AHP Framework Corrects This

The AHP model corrects the imbalance by explicitly prioritising Social factors and embedding product-related externalities into the weighing structure. When applied to the data analysed, this produces a materially lower adjusted ESG score than reported in ratings, widening the EMI gap over time. In fact, the BAT-adjusted AHP index for 2025 comes to 39.0, while the reported composite score is 81.0. The difference of 42 points makes the EMI figure, and this figure has been increasing annually to 39.5 points in 2023, 40.8 points in 2024, and 42.0 points in 2025. Importantly, this is not unique to BAT: peer companies



converge under the same framework, reflecting a shared product-risk profile. The value of AHP thus lies in exposing a systematic bias in mainstream ESG methodologies, namely the under-penalisation of product harm, as also shown in the sub-criterion scores: S1 = 20/100, S2 = 35/100, S3 = 38/100, and E1 = 45/100.

5.3. Does "Better Tomorrow" Hold Up?

The Royal College of Physicians (2025) estimates vapour products carry under 5% of combustible risk for adult users. However, it measures personal risk, not population harm. In England, 18-24 year-old vaping rose from 5.0% to 25.0% over 10 years. Among those who never smoked, the vaping rate within that age group increased to 34.3% (Jackson et al.). US CDC data shows e-cigarette use at 11.0% among 18-24 year olds versus 4.5% across all adults (Cornelius et al., 2022). APMEA revenues fell 10.86% while AME revenues rose, confirming geographic harm substitution rather than net harm reduction.

5.4. Gap Analysis and Limitations

The ESG gap reflects structural limitations in conventional rating methodologies, which currently prioritise operational metrics (such as Scope 1 and 2 emissions reductions) over product-related externalities and Scope 3 effects. Additionally, Scope 3 emissions and downstream impacts are often insufficiently incorporated, further distorting the overall assessment. AHP introduces subjectivity, sensitivity analysis shows score variations of only ± 3 -5 points under different assumptions, without altering the direction of results. The EMI gap hence remains both significant and widening, supporting the case for sector-specific ESG approaches.

6. Discussion & Policy Implications

6.1. Main Findings

The EMI results show that BAT's reported ESG performance improves steadily over time, but the materially adjusted assessment improves much more slowly, which creates a widening gap between image and impact. In 2023, the reported ESG score stands at 74.5, while the AHP-adjusted score is only 35.0; by 2025, these values rise to 81.0 and 39.0 respectively, leaving an EMI gap of 42.0 points. This pattern suggests that BAT is making progress in ways that are visible to ESG rating systems, but that these gains are not translating into equally strong reductions in the harms that matter most. The Social pillar is the clearest source of concern, because youth vaping, dependency, and geographic harm transfer continue to weigh heavily in the adjusted framework. In short, the EMI does not suggest that BAT has made no progress; rather, it shows that the progress is modest when judged against the full scale of its social and health impact.

6.2. Policy Implications

Three groups face direct implications from these findings. Regulators should require independent verification of harm-reduction product claims before any ESG classification is granted, and WHO FCTC ratification gaps enabling a regulatory vacuum in Sub-Saharan Africa warrant multilateral attention. Investors should treat the divergence across FTSE4Good, Sustainalytics, and MSCI as a signal of greenwashing exposure rather than analytical disagreement. GRI and SASB should introduce a user-recruitment metric for tobacco that separates smoker conversion from market expansion, and move toward sector-specific harm-adjusted weighting for controversial industries.

7. Conclusion

Overall, the report finds that BAT's ESG ratings are improving, but not in a way that convincingly reflects a comparable improvement in real-world impact. The widening EMI gap from 39.5 in 2023 to 42.0 in 2025 shows that reported sustainability progress is



outpacing materially grounded progress, and suggests that current ESG systems may be giving BAT more credit for managing perceptions than for reducing harm. The strongest conclusion is that tobacco should not be judged by operational improvements alone: if the product continues to recruit users, sustain dependency, and shift harm across regions, then ESG success remains incomplete. In that sense, the report's contribution is both analytical and practical; it offers a way to measure misrepresentation and a reminder that genuine sustainability in tobacco must be judged by reduced harm, not improved branding.

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